



```
$ docker compose up -d --build # SIMKODES ENTERPRISE DEVOPS
```

# SIMKODES: Cloud DevOps & Infrastructure Pipeline

Arsitektur Multi-Container Docker (PHP 8.2-FPM, Nginx, MySQL), Hostinger Cloud Deployment & CI/CD

# Peta Jalan & Sistematika DevOps SIMKODES

Tahapan implementasi containerization, hosting cloud deployment, dan otomatisasi pipeline

01

## Live Environment & Interface

Showcase antarmuka produksi SIMKODES di atas runtime PHP 8.2-FPM

02

## Cloud Hosting & Control Panel

Konfigurasi server Hostinger, web root, dan manajemen resource database

03

## Docker Multi-Container Architecture

Spesifikasi Dockerfile, docker-compose orchestration, dan Nginx gateway

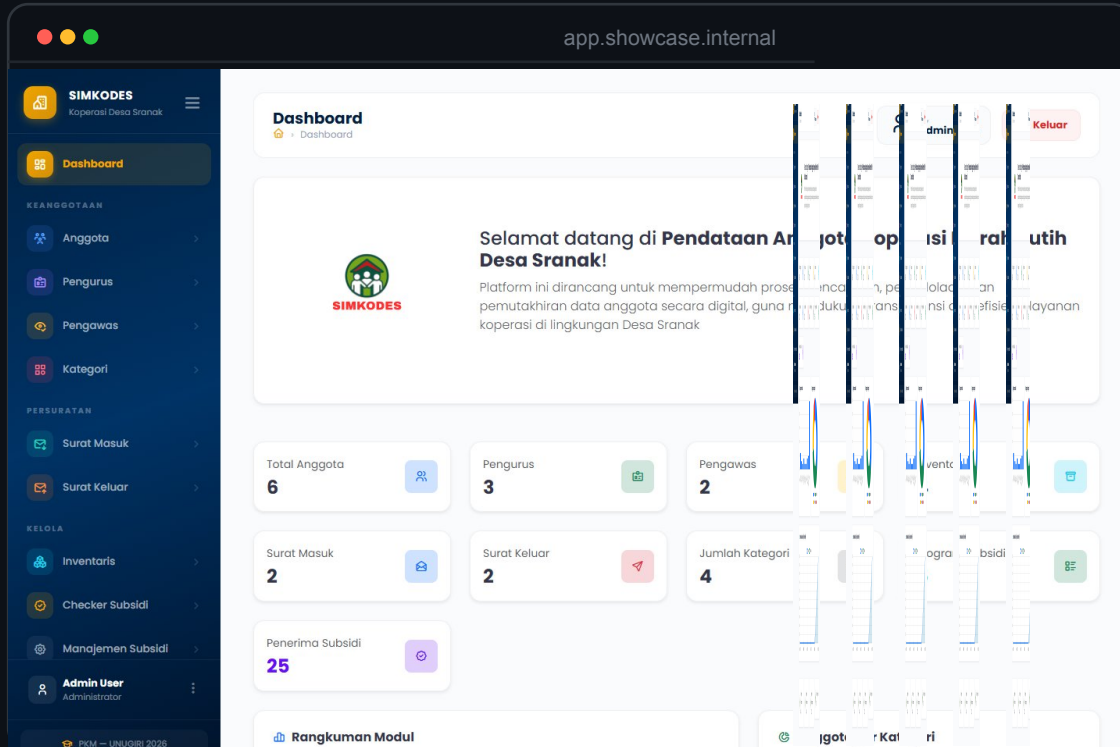
04

## CI/CD & Deployment Automation

Automated deployment pipeline, zero-downtime release, dan cache optimization

# Aplikasi SIMKODES Live (Production)

Dashboard operasional yang berjalan di atas container Docker & PHP 8.2-FPM

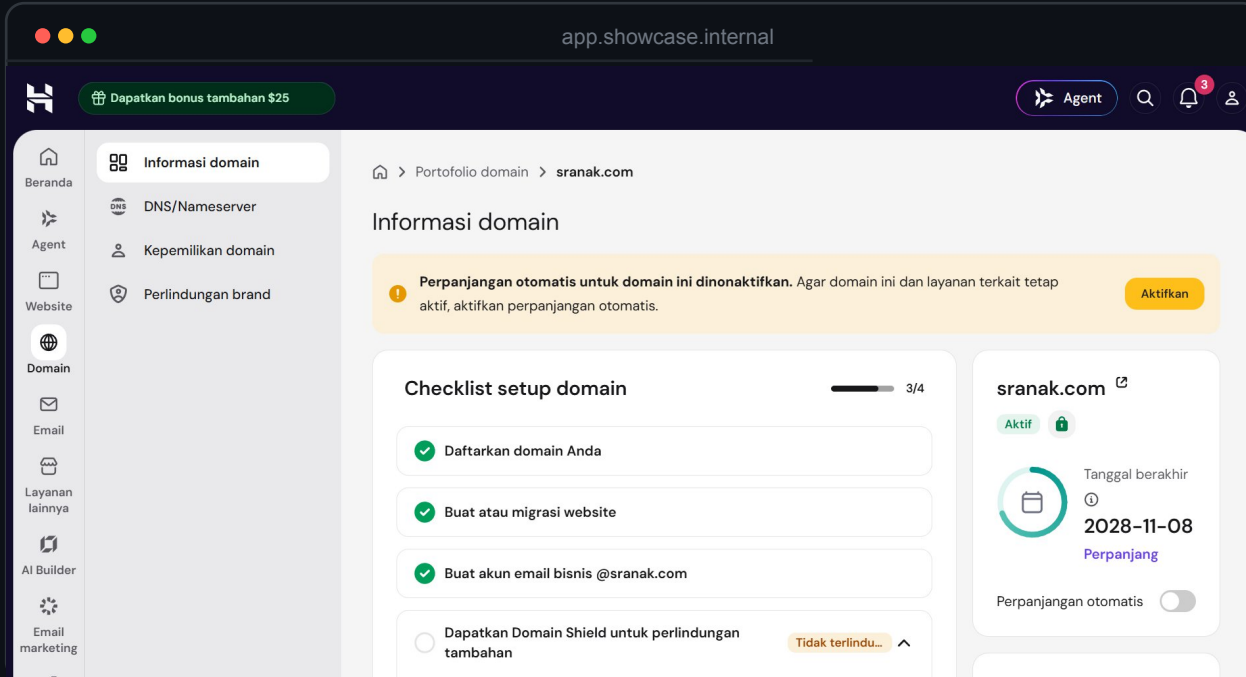


## FITUR & KOMPONEN UTAMA:

- \$ > Runtime PHP 8.2-FPM
- terisolasi dalam container Docker
- \$ > Query SQL teroptimasi dengan database indexing dan OPcache
- \$ > Asset bundling frontend modern menggunakan Vite compiler

# Infrastruktur Server Cloud Hosting

Konfigurasi domain routing, web root public, dan SSL/TLS certificate

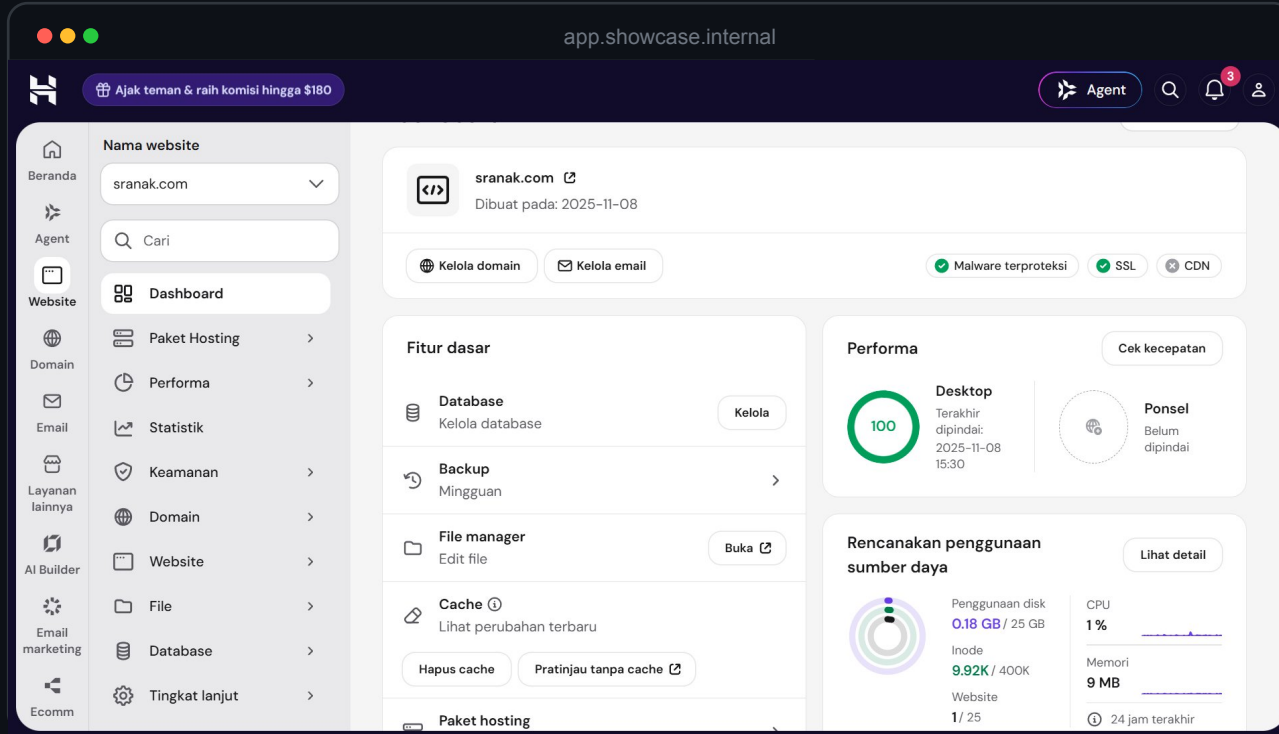


## FITUR & KOMPONEN UTAMA:

- \$ > Web root terarah ke direktori /public untuk keamanan file
- \$ > Konfigurasi FastCGI gateway dan custom environment PHP
- \$ > Integrasi SSL/TLS otomatis untuk enkripsi data traffic HTTPS

# Panel Kontrol Hosting & Database

Monitoring alokasi resource CPU, RAM, disk storage, dan database MySQL



## FITUR & KOMPONEN UTAMA:

- \$ > Database MySQL 8.0 dengan isolasi kredensial dan hak akses aman
- \$ > Monitoring performa memori dan CPU usage secara real-time
- \$ > Automated backup snapshot berkala untuk disaster recovery

# Arsitektur Ekosistem DevOps & Multi-Container

Tiga pilar utama penggerak performa, isolasi service, & keandalan infrastruktur



Application  
Layer

## App Engine (PHP 8.2-FPM)

Environment aplikasi terisolasi dilengkapi ekstensi GD, Zip, Intl, dan OPcache bytecode caching.



Gateway  
Layer

## Nginx Edge Proxy

Web server reverse proxy dengan FastCGI pass-through, gzip compression, dan security headers.



Storage  
Layer

## MySQL 8.0 & Persistence

Database relasional dengan Docker volume persisten untuk menjamin keutuhan data transaksi.

# Spesifikasi Dockerfile: PHP 8.2 Engine

DOCKERFILE

```
$ docker build -t simkodes-app:latest .
```

• ACTIVE

```
FROM php:8.2-fpm
```

```
# System dependencies & PHP extensions
```

```
RUN apt-get update && apt-get install -y \  
    git curl libpng-dev libzip-dev libicu-dev unzip \  
    && docker-php-ext-configure gd --with-freetype --with-jpeg \  
    && docker-php-ext-install -j$(nproc) \  
    pdo_mysql mbstring exif pcntl bcmath gd zip intl opcache
```

```
# Composer binary from official image
```

```
COPY --from=composer:latest /usr/bin/composer /usr/bin/composer
```

```
WORKDIR /var/www/html
```

```
COPY . /var/www/html
```

```
COPY docker/entrypoint.sh /usr/local/bin/entrypoint.sh
```

```
RUN chmod +x /usr/local/bin/entrypoint.sh
```

```
ENTRYPOINT ["/usr/local/bin/entrypoint.sh"]
```

```
CMD ["php-fpm"]
```

# Orkestrasi Docker Compose Multi-Container

YAML

```
$ docker compose up -d
```

• ACTIVE

```
services:
  app:
    build: .
    container_name: simkodes_app
    volumes:
      - ./var/www/html
      - ./docker/php/php.ini:/usr/local/etc/php/conf.d/custom.ini:ro
    networks: [simkodes_network]
    depends_on: [db]

  nginx:
    image: nginx:alpine
    container_name: simkodes_nginx
    ports: ["8000:80"]
    volumes:
      - ./var/www/html
      - ./docker/nginx/default.conf:/etc/nginx/conf.d/default.conf:ro
    networks: [simkodes_network]
    depends_on: [app]

  db:
    image: mysql:8.0
    container_name: simkodes_db
    volumes: [simkodes_db_data:/var/lib/mysql]
    networks: [simkodes_network]
```

# Automasi Deployment Script: hosting-setup.sh

BASH

\$ bash hosting-setup.sh

• ACTIVE

```
#!/usr/bin/env bash
set -euo pipefail
```

```
echo "==> 1. Inisialisasi Environment & Composer"
composer install --no-dev --optimize-autoloader
```

```
echo "==> 2. Generate Application Security Key"
php artisan key:generate --force
```

```
echo "==> 3. Database Migration & Storage Link"
php artisan migrate --force
php artisan storage:link
```

```
echo "==> 4. Build Vite Assets & Cache Optimization"
npm install && npm run build
php artisan config:cache && php artisan route:cache && php artisan view:cache
```

```
echo "==> Deployment Sukses & Sistem Siap Digunakan!"
```

# Terima Kasih

SIMKODES Cloud DevOps & Containerized Infrastructure Pipeline

2026